

$$\text{hinge loss} = \begin{cases} 1 - yx & \text{if } yx < 1 \\ 0 & \text{if } yx \geq 1 \end{cases}$$

$$\max_x (y - yx)$$

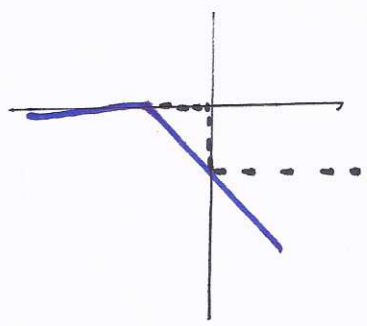
$$\lambda = \frac{\partial}{\partial x} \sum_{n=1}^N \mathcal{E}(y_n) + \lambda \|W\|^2$$

$$\frac{\partial}{\partial x} \left(\sum_{n=1}^N \mathcal{E}(y_n) + \lambda \|W\|^2 \right)$$

$$\sum_{n=1}^N \mathcal{E}(y_n) + \lambda \|W\|^2$$

$$\sum_{n=1}^N \mathcal{E}(y_n) + \lambda \|W\|^2$$

hinge loss



$$\left\{ \begin{aligned} \frac{\partial \mathcal{E}}{\partial w_0} &= \sum_{i=1}^n (w_0 + w_1 x_i - y_i) = 0 \\ \frac{\partial \mathcal{E}}{\partial w_1} &= \sum_{i=1}^n (w_0 + w_1 x_i - y_i) x_i = 0 \end{aligned} \right.$$

$$\begin{aligned} \text{Error}(w) &= \frac{1}{2} \sum_{i=1}^n (h_w(x_i) - y_i)^2 + \frac{\lambda}{2} \|w\|^2 \\ h(w) &= w_0 + w_1 x \\ E(w) &= \frac{1}{2} \sum_{i=1}^n (w_0 + w_1 x_i - y_i)^2 + \frac{\lambda}{2} (w_0^2 + w_1^2) \end{aligned}$$

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